THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 2: **Polynomials** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	$0x^5$ and $0x^3$ vanish; highest power with non-zero coefficient is 4
2	**(a)** $5, -2$	$x^2 - 3x - 10 = (x-5)(x+2)$
3	**(a)** 0	Graph never meets the x-axis ⇒ no zero
4	**(b)** −5	$\alpha + \beta = -\frac{b}{a} = -5$
5	**(b)** $x^2 + 3x + 2$	$x^2 - (\text{sum})x + \text{product} = x^2 + 3x + 2$
6	**(b)** Incorrect — it can have at most 2 zeroes	A polynomial of degree n has at most n zeroes
7	**(a)** 5	$4 - 2k + 6 = 0 \Rightarrow k = 5$
8	**(a)** Student A is correct	$(x-4)(x+4)$ ⇒ $4, -4$
9	**(b)** $-\frac{2}{3}$	$\alpha\beta = \frac{c}{a} = -\frac{2}{3}$
10	**(b)** $b = 0$	Sum $= -\frac{b}{a} = 0 \Rightarrow b = 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$4u(u+2) = 0 \Rightarrow u = 0, -2$. Sum $= -2 = -\frac{8}{4}$ ✓; product $= 0 = \frac{0}{4}$ ✓	1+1
12	$x^2 - \frac{1}{4}x - 1$, i.e. $k(4x^2 - x - 4)$; e.g. $4x^2 - x - 4$	2
13	$x^2 + 1 \geq 1 > 0$ for every real x, so it never becomes 0; its graph does not meet the x-axis. It has **no real zero** — the claim is wrong.	1+1
14	$\alpha + \beta = 6$, $\alpha\beta = 8$; $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 36 - 16 = 20$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$6x^2 - 7x - 3 = (3x+1)(2x-3)$ ⇒ zeroes $-\frac{1}{3}$, $\frac{3}{2}$. Sum $= \frac{7}{6} = -\frac{-7}{6}$ ✓; product $= -\frac{1}{2} = \frac{-3}{6}$ ✓	1+1+1
16	$\alpha + \beta = \frac{5}{2}$, $\alpha\beta = \frac{3}{2}$. (i) $\frac{\alpha+\beta}{\alpha\beta} = \frac{5}{3}$ (ii) $\alpha\beta(\alpha+\beta) = \frac{15}{4}$ (iii) $(\alpha+\beta)^2 - 4\alpha\beta = \frac{25}{4} - 6 = \frac{1}{4}$	1+1+1
17	Sum $= 6$, product $= 9 - 2 = 7 \Rightarrow x^2 - 6x + 7$. Check: $-\frac{b}{a} = 6$, $\frac{c}{a} = 7$ ✓	1+1+1
	OR: Let zeroes be $\alpha, 7\alpha$: $8\alpha = \frac{8}{3} \Rightarrow \alpha = \frac{1}{3}$; zeroes $\frac{1}{3}, \frac{7}{3}$. Product $\frac{7}{9} = \frac{2k+1}{3} \Rightarrow k = \frac{2}{3}$	1+1+1
18	Table: $x = -3, -2, -1, 0, 1, 2, 3, 4$ give $y = 6, 0, -4, -6, -6, -4, 0, 6$. The parabola cuts the x-axis at $(-2, 0)$ and $(3, 0)$ ⇒ zeroes -2 and 3.	2+1
	OR: $p(1) = 0$, $p(-1) = 0$, $p(-3) = -27 + 27 + 3 - 3 = 0$ ✓. Sum $= -3 = -\frac{b}{a}$; $\Sigma\alpha\beta = -1 + 3 - 3 = -1 = \frac{c}{a}$; product $= 3 = -\frac{d}{a}$ ✓	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$(3x-4)(x+1)$ ⇒ zeroes $\frac{4}{3}, -1$. Sum $\frac{1}{3} = -\frac{-1}{3}$ ✓, product $-\frac{4}{3}$ ✓ (3 marks). Reciprocals $\frac{3}{4}, -1$: sum $-\frac{1}{4}$, product $-\frac{3}{4}$ ⇒ $4x^2 + x - 3$ (2 marks).	3+2
	OR: (i) $-(x^2 - 4x - 5) = -(x-5)(x+1)$ ⇒ $5, -1$. (ii) $x = 5$: the ball lands 5 m away (-1 is rejected). (iii) Sum $4 = -\frac{4}{-1}$ ✓; product $-5 = \frac{5}{-1}$ ✓	2+1+2
20	(i) 3 zeroes: $-2, 1, 3$ (ii) degree 3 (iii) $(x+2)(x-1)(x-3) = x^3 - 2x^2 - 5x + 6$ (iv) $-2 + 1 + 3 = 2 = -\frac{-2}{1}$ ✓	1+1+2+1
	OR: $(\alpha - \beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta \Rightarrow 1 = 25 - 4k \Rightarrow k = 6$; zeroes 3 and 2 (3 marks). Zeroes $6, 4$: sum 10, product 24 ⇒ $x^2 - 10x + 24$ (2 marks).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 2: **Polynomials** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $2x^2 - \sqrt{3}x + 1$	Degree 2 with non-negative integral powers
2	**(b)** 1	Number of distinct points on x-axis = number of zeroes = 1
3	**(b)** $3, -3$	$(x-3)(x+3)$
4	**(b)** 4	$-\frac{-8}{2} = 4$
5	**(c)** 12	$\frac{c}{a} = 12$
6	**(b)** Incorrect; zeroes are -2 and -5	$(x+2)(x+5) \Rightarrow -2, -5$
7	**(b)** −3	$1 - k - 4 = 0 \Rightarrow k = -3$
8	**(a)** Student A is correct	Sum -1, product $-6 \Rightarrow x^2 + x - 6$
9	**(c)** 3	Degree 3 ⇒ at most 3 zeroes
10	**(c)** $c = 0$	$p(0) = c = 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(2s-1)^2 \Rightarrow$ zeroes $\frac{1}{2}, \frac{1}{2}$. Sum $1 = \frac{4}{4}$ ✓; product $\frac{1}{4}$ ✓	1+1
12	$x^2 - \sqrt{2}x + \frac{1}{3}$, i.e. $k(3x^2 - 3\sqrt{2}x + 1)$	2
13	$\alpha + \beta = 5,\ \alpha\beta = 6 \Rightarrow \frac{\alpha+\beta}{\alpha\beta} = \frac{5}{6}$	1+1
14	$x^2 - 4x + 4 = (x-2)^2 \Rightarrow$ both zeroes equal 2; the graph only touches the x-axis at $(2, 0)$. Claim is **false**.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$(x - \sqrt{3})(x + \sqrt{3}) \Rightarrow \pm\sqrt{3}$. Sum $0 = -\frac{0}{1}$ ✓; product $-3 = \frac{-3}{1}$ ✓	1+1+1
16	$\alpha + \beta = 1,\ \alpha\beta = -2$. New sum $= 2(1) + 2 = 4$; product $= 4\alpha\beta + 2(\alpha + \beta) + 1 = -8 + 2 + 1 = -5 \Rightarrow x^2 - 4x - 5$	1+1+1
17	Sum $-1 = -(a+1) \Rightarrow a = 0$; product $-6 = b$. So $a = 0,\ b = -6$	1½+1½
	OR: $-\frac{2}{k} = \frac{3k}{k} = 3 \Rightarrow k = -\frac{2}{3}$. Polynomial: $-\frac{2}{3}x^2 + 2x - 2$, i.e. $-\frac{2}{3}(x^2 - 3x + 3)$	2+1
18	$x^3 - (\text{sum})x^2 + (\Sigma\alpha\beta)x - \text{product} = x^3 - 2x^2 - 7x + 14$	3
	OR: $p(5) = 375 - 250 - 135 + 10 = 0$; $p(-2) = -24 - 40 + 54 + 10 = 0$; $p(\frac{1}{3}) = \frac{1}{9} - \frac{10}{9} - 9 + 10 = 0$ ✓. Sum $= \frac{10}{3} = -\frac{-10}{3}$ ✓	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(2x+3)(x-2) \Rightarrow 2, -\frac{3}{2}$; sum $\frac{1}{2} = -\frac{-1}{2}$ ✓, product $-3 = \frac{-6}{2}$ ✓. (ii) Zeroes $4, -3$: $x^2 - x - 12$	3+2
	OR: Table: $x = -2, -1, 0, 1, 2, 3, 4 \to y = 5, 0, -3, -4, -3, 0, 5$ (2). Graph cuts x-axis at -1 and 3 (2). Sum $2 = -\frac{-2}{1}$, product -3 ✓ (1)	2+2+1
20	(i) $-(x-1)(x-5) \Rightarrow x = 1, 5$ (ii) $5 - 1 = 4$ m (iii) sum $6 = -\frac{6}{-1}$, product $5 = \frac{-5}{-1}$ ✓ (iv) $-9 + 18 - 5 = 4$ m	2+1+1+1
	OR: $\alpha + \beta = 4,\ \alpha\beta = 3$. $\alpha^2 + \beta^2 = 16 - 6 = 10$, $\alpha^2\beta^2 = 9 \Rightarrow x^2 - 10x + 9$ (3). $\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = 64 - 36 = 28$ (2)	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.